AI Coding with Junie

JetBrains' Coding Agent for IDE, CLI, and CI

Plan, implement, review, and verify
with a coding agent that understands your project

🎯 What You'll Master Today

Core Concepts

- 🔍 **Ask Mode**: Read-only exploration
- 🧭 **Plan Mode**: Align before coding
- ⚡ **Code Mode**: Implement and verify
- 🛡️ **Safety**: Approvals, allowlists, review

Hands-On Labs

- ☕ **Java**: Spring Boot APIs
- 🐍 **Python**: Refactoring & testing
- ⚛️ **React**: Forms & validation
- 🔧 **MCP**: Current docs and tools

📚 Prerequisites Check

✅ **JetBrains IDE** (IntelliJ IDEA, PyCharm, WebStorm, GoLand, PhpStorm, RustRover, RubyMine)

✅ **JetBrains AI** subscription, Junie API key, or BYOK provider key

✅ **Development Environment** (Java 21+ / Python 3.8+ / Node 18+)

Cost control: Use `/usage`, choose models intentionally, and prefer BYOK/local models for heavy practice

💡 Quick Setup: We'll verify Junie, CLI, and lab runtimes together at the start

🚀 Getting Started — Two Surfaces

JetBrains IDE

1. AI Chat → Agent dropdown → **Junie**
2. Separate Junie tool window remains available
3. Best for indexes, refactorings, debugger, inspections

Junie CLI

1. `brew tap jetbrains-junie/junie && brew install junie`
2. Or: `curl -fsSL https://junie.jetbrains.com/install.sh | bash`
3. Run `junie` in any project, or `junie "task"` headlessly

Model and provider options:

JetBrains account / credits

OpenAI / Anthropic / Google / xAI

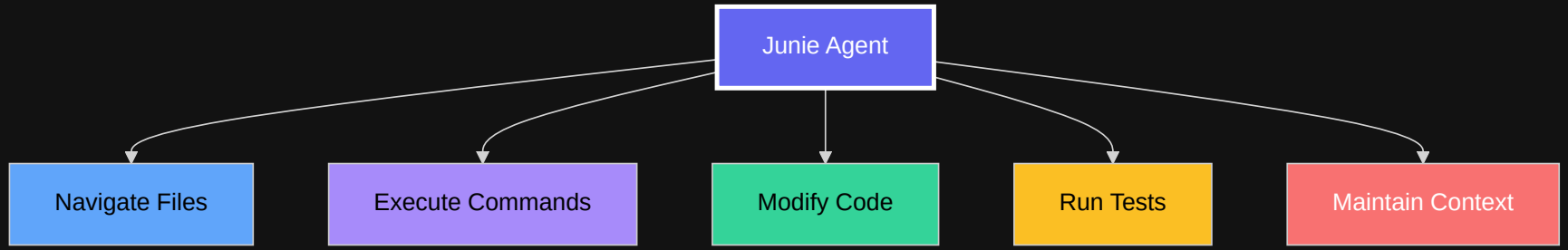
OpenRouter / Copilot

Ollama / LM Studio / LiteLLM

Use `/model` and `/usage` to stay intentional

🤖 What is Junie?

A coding agent that works where JetBrains developers already work



🆕 Junie After Beta

General Availability Story

- JetBrains now calls Junie a coding agent
- IDE, terminal, and CI/CD are one product story
- Plans before it codes
- Runs long tasks while you review intent

Plan Mode Is Central

- Read-only exploration first
- Requirements, design, tests, delivery steps
- Review and refine before implementation
- Save plans as durable task docs

🆕 Junie After Beta (continued)

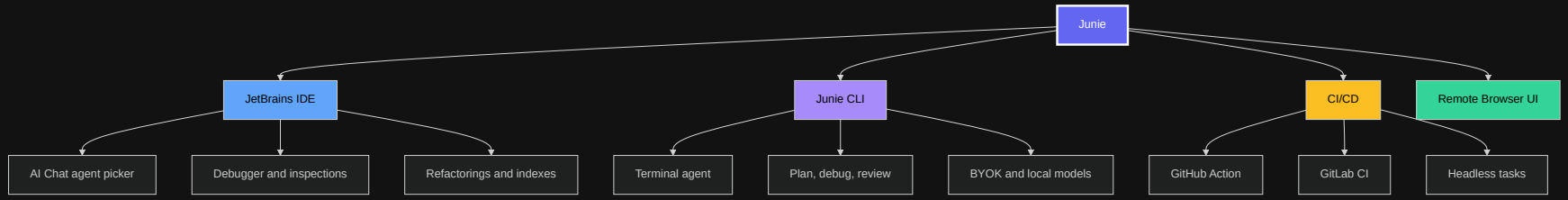
Shared Agent Context

- `.junie/AGENTS.md` first
- Root `AGENTS.md` also supported
- Skills live in `.junie/skills/`
- Extensions bundle skills, MCP, commands, agents

Beyond the IDE

- `/review` for local diffs
- `/debug` with JetBrains debugger context
- `/remote` to continue a CLI session in browser
- GitHub Action and GitLab CI workflows

Junie Surfaces



The CLI can connect to a matching JetBrains IDE for symbol-aware search, inspections, test discovery, and debugger-backed workflows.

🔄 Three Working Modes

🔍 Ask Mode

Read-Only Analysis

- Explain complex code
- Analyze architecture
- Find bugs & issues
- Review security
- Check test coverage

Perfect for understanding before changing

🧭 Plan Mode

Design Before Edits

- Requirements
- Technical design
- Testing strategy
- Delivery steps
- Confirm or refine

Best for non-trivial tasks

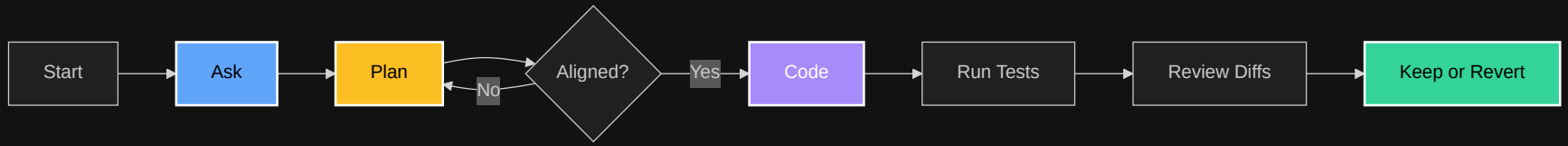
⚡ Code Mode

Make Changes

- Implement features
- Fix bugs
- Refactor code
- Generate tests
- Update dependencies

Executes multi-step plans with diffs

💡 Professional Workflow



Use Plan mode when the cost of the wrong implementation is higher than the cost of five minutes of alignment.

🧭 Plan Mode in Practice

Start It

- Press `Shift+Tab` in Junie CLI
- Or type `/plan <task>`
- In the IDE, start with Ask/Auto and request a plan first
- For installed flags, check `junie --help`

Review It

- Read assumptions
- Edit scope before code exists
- Check testing strategy
- Confirm implementation only after the plan matches your intent

Teaching move: use one complete prompt, let Junie plan, then ask attendees what they would change before implementation.

📝 Project Guidelines

Preferred: `.junie/AGENTS.md` | **Also supported:** root `AGENTS.md` | **Legacy:** `.junie/guidelines.md`

```
## Technology Stack
- Framework: Spring Boot 3.2
- Testing: JUnit 5 + AssertJ

## Conventions
- REST: /api/v1/{resource}
- DTOs: Java records
- Tests: Given-When-Then
```

Lookup order: `.junie/AGENTS.md` → `AGENTS.md` (project root) → `.junie/guidelines.md` (legacy). Global guidelines can live in `~/.junie/AGENTS.md`.

❌ Without Guidelines

Inconsistent patterns

Multiple review cycles

Style conflicts

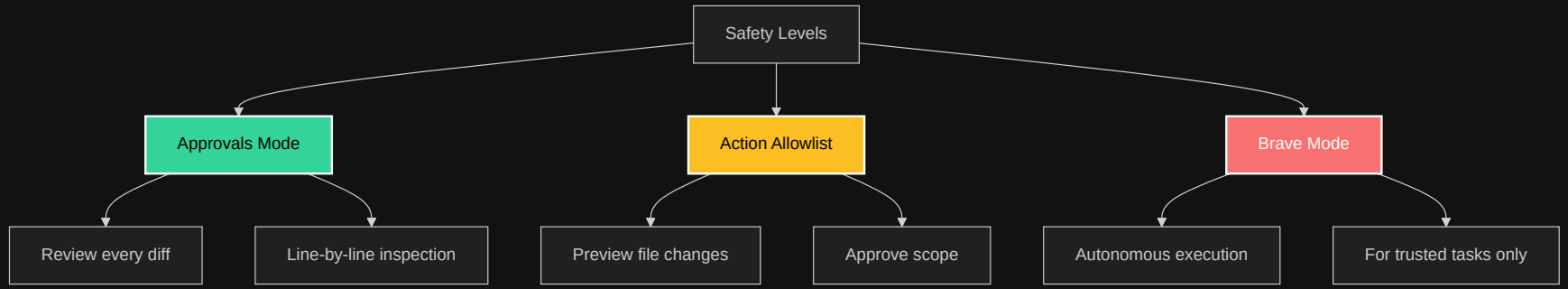
✅ With Guidelines

Consistent output

Fewer review cycles

Team alignment

Safety Controls



⚡ When to Use Each Mode

Mode	Good For	Not For
Approvals	• Critical business logic • First-time tasks • Learning Junie	• Repetitive formatting • Simple refactors
Allowlist	• Known file sets • Defined scope • Team reviews	• Exploratory changes • Unknown impact
Brave	• Test generation • Formatting • Documentation	• Production code • Database changes • First attempts

🔌 MCP: Model Context Protocol

Use MCP For

- External services
- Current docs
- Browser exploration
- Shared project tools

Use Skills For

- Repeatable workflows
- Test conventions
- Review checklists
- Team patterns

```
{
  "mcpServers": {
    "context7": {
      "command": "npx",
      "args": ["-y", "@upstash/context7-mcp"]
    },
    "playwright": {
      "command": "npx",
      "args": ["-y", "@playwright/mcp"]
    }
  }
}
```

Where it lives

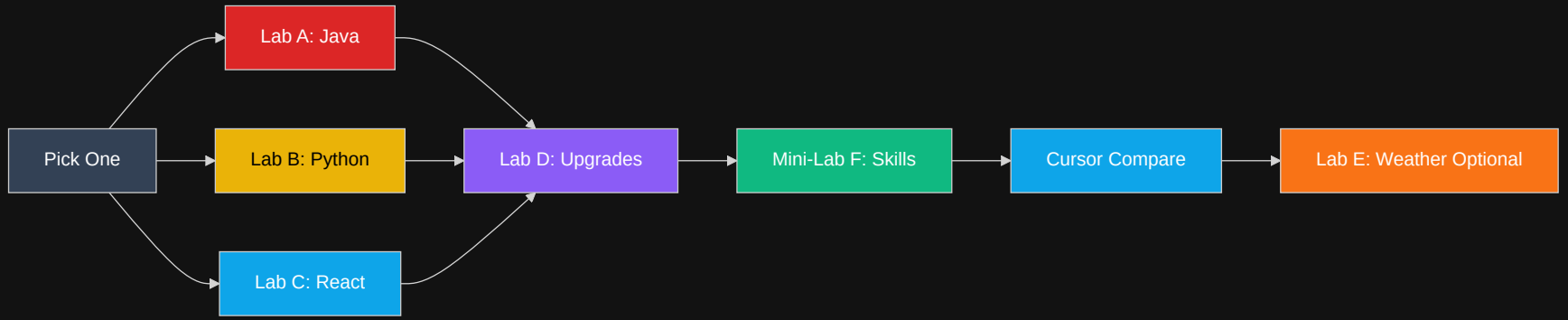
CLI: `/mcp`

Project: `.junie/mcp/mcp.json`

User: `~/.junie/mcp/mcp.json`

context7 is the main workshop MCP example.

🧪 Lab Overview



🧪 Lab Overview — Tracks

☕ Java
Spring Boot
REST APIs
JUnit + AssertJ

🐍 Python
PEP 8
Type hints
pytest

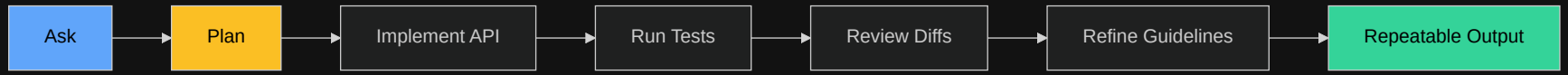
⚛️ React
TypeScript
Forms
Testing Library

🔧 MCP
context7
Upgrades
Migration

🧩 Skills
Reusable
Workflow
Playwright

🌦️ Lab E
Weather API
Proxy
Optional

☕ Lab A: Spring Boot Journey



Baseline Pass

- Ask Junie to inspect first
- Let it propose a plan
- Watch where style drifts

Guided Pass

- `.junie/AGENTS.md` loaded
- One complete feature prompt
- Tests and review included

🐍 Lab B: Python Transformation

Starting Point

```python
def calc(x,y,op):
    if op=="add": return x+y
    elif op=="sub": return x-y
```

- No type hints
- Poor naming
- No tests
- No docs

Target Outcome

```python
def calculate(
    left: float,
    right: float,
    operation: Operation,
) -> float:
    """Apply a supported operation."""
```

- Type-safe API
- pytest coverage
- Documented edge cases

⚛️ Lab C: React Forms

```
interface RegistrationForm {
  email: string;      // valid email
  password: string;   // min 8, special char
  confirm: string;    // must match
  terms: boolean;     // required
}
```

React Hook Form
Form state

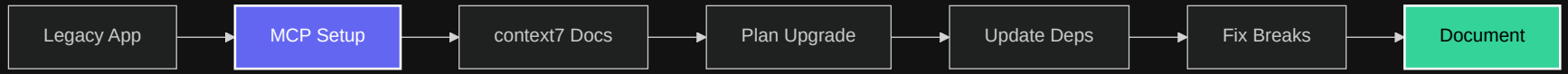
Zod
Validation

Testing Library
90% coverage

Accessibility
ARIA compliant

Teaching move: Use one complete prompt that asks for the component, validation, accessible errors, tests, and a final self-review.

🔧 Lab D: Smart Upgrades with context7



React 17 → React 18

Legacy → Modern

Four-hour version: emphasize MCP setup, migration plan quality, and one contained implementation slice.

🧩 Mini-Lab: Create a Playwright Skill

Capture the workflow once, reuse it every time

```
Create a skill named playwright-e2e that captures our
browser testing conventions:
- Prefer accessible locators
- Use page objects only after duplication appears
- Run npx playwright test
- Summarize failures and screenshots
```

1. Inspect Existing Skills
2. Ask Junie to Create One
3. Use It on a Test
4. Run Local CLI

Teaching point: Playwright MCP is good for interactive browser exploration. A Skill is better for repeatable team testing conventions.

🎯 Common Patterns

TDD Flow

1. Ask: "What should fail?"
2. Plan: "How will we prove it?"
3. Code: "Implement and run tests"

Red → Green → Refactor

Plan-First Feature

1. State acceptance criteria
2. Review Junie's plan
3. Approve a focused diff

Scope stays visible

Upgrades

1. Use current docs
2. Plan migration
3. Verify behavior

Systematic migration

🚀 CI/CD Integration — Junie GitHub Action

```yaml
name: Junie Agent
on:
  issue_comment:
    types: [created]
  pull_request_review_comment:
    types: [created]

permissions: { contents: write, pull-requests: write, issues: write }

jobs:
  junie:
    if: contains(github.event.comment.body, '@junie-agent')
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - uses: JetBrains/junie-github-action@v1
        with:
          junie_api_key: ${{ secrets.JUNIE_API_KEY }}
```

GitHub: `@junie-agent` in issues or PRs

GitLab: `#junie` in merge request comments

CLI: `junie "prompt"` for headless local automation

📱 Remote Mode Demo

What It Is

- Browser UI for the same CLI session
- Works from laptop, tablet, or phone browser
- Terminal and web UI stay synchronized
- Your machine must stay awake

Demo Flow

1. Start `junie` in the terminal
2. Run `/remote`
3. Open `junie.jetbrains.com/remote`
4. Respond from the browser
5. Run `/remote` and choose Stop Remote Session

Remote mode uses the Junie service. JetBrains Account sign-in is the simplest path; a Junie API key is the alternate route.

🤔 Why Junie Instead of Cursor?

Use Junie When…

- Team lives in JetBrains IDEs
- IDE refactorings matter
- Plans before edits matter
- IDE, CLI, and CI should align

Use Cursor When…

- AI-first editor is desired
- Cursor Tab is central
- Background agents matter
- Team already uses rules/Bugbot

Compare Live

- Same repo
- Same task
- Same acceptance criteria
- Stop before a second workshop

Cursor can also run inside JetBrains through ACP, so this is about workflow fit.

✨ Key Takeaways

✅ **Ask before Code** - Understand first, implement second

✅ **Plan before big changes** - Align on requirements, tests, and scope

✅ **Guidelines drive consistency** - Encode your team's DNA

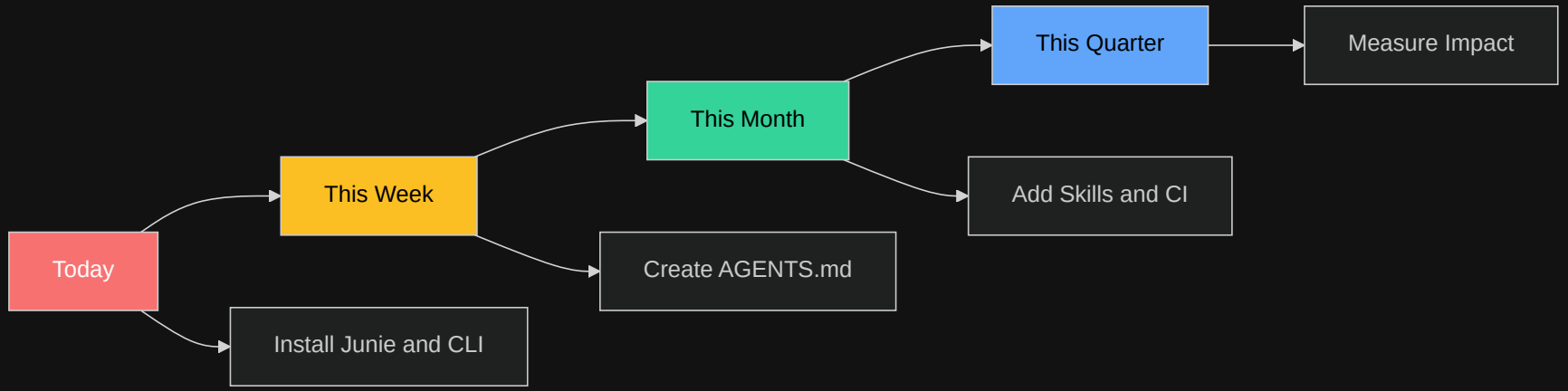
✅ **Skills preserve workflows** - Reuse task-specific patterns

✅ **Build trust gradually** - Approvals → Allowlist → Brave

✅ **CLI matters** - Use Junie outside the IDE and in automation

✅ **Safety first** - Review diffs, run tests, maintain control

🎯 Your Next Steps



🚀 Start with Ask and Plan → Approve focused diffs → Add guidelines → Automate later

💬 Discussion Time

Let’s Explore Together:

- 🤔 **IDE plugin vs CLI vs CI/CD — which fits your workflow?**
- 🛡️ **Where do you always want human review?**
- 📋 **How could AGENTS.md guidelines help your team?**
- 🔧 **Which MCP tools and Agent Skills would you create?**
- 🚀 **Where does Cursor, Codex, Claude, or Antigravity fit better?**

📚 Resources Hub

Documentation

- 📖 Junie Docs
- 📝 Guidelines & Memory
- 🔧 Agent Skills
- 🖥️ CLI Reference

Agent Workflows

- 🐙 GitHub Action
- 🦊 GitLab CI
- 🧭 Plan Mode
- 📱 Remote Mode

GitHub

- 🗂️ Guidelines Catalog
- 📚 Context7 MCP
- 🧪 This repository

Adjacent Tools

- 🖊️ Cursor Docs
- 🤖 Codex / Claude / Antigravity docs
- 🔌 Agent Communication Protocol

Community

- 💬 JetBrains AI Slack
- 🏷️ Stack Overflow: `jetbrains-junie`
- 📧 Support team

🙏 Thank You!

AI amplifies good practices

💡 Remember:
Good agents amplify clear intent, tested code, and team conventions.

Questions? Let's explore together! 🚀

About Ken Kousen

Ken Kousen

President, Kousen IT, Inc.

📧 ken.kousen@kousenit.com
🐙 github.com/kousen
📺 @talesfromthejarside
📝 kousenit.substack.com
💼 linkedin.com/in/kenkousen
🦋 bsky.app/profile/kousenit.com

AI agents aren't here to replace developers—
they're here to make us **better** developers.

🎁 Bonus: Quick Reference

IDE Shortcuts

• `Ctrl+Alt+J` - Open Junie
• `Ctrl+Enter` - Execute
• `Esc` - Cancel operation ### CLI Shortcuts • `Shift+Tab` - Plan mode
• `/mcp` - MCP setup
• `/model` - Choose model
• `/effort` - Reasoning effort/cost
• `/usage` - Credits usage
• `Ctrl+B` - Brave mode
• `Ctrl+R` - Prompt history

Common Commands

"Analyze this file"
"Create a plan first"
"Add tests with 90% coverage"
"Implement and run tests"
"Review the diff"

MCP Tools

```
{
  "context7": "Library docs",
  "playwright": "E2E tests",
  "search": "Web search",
  "custom": "Your tools"
}
```

🎯 Print this slide for your desk!